



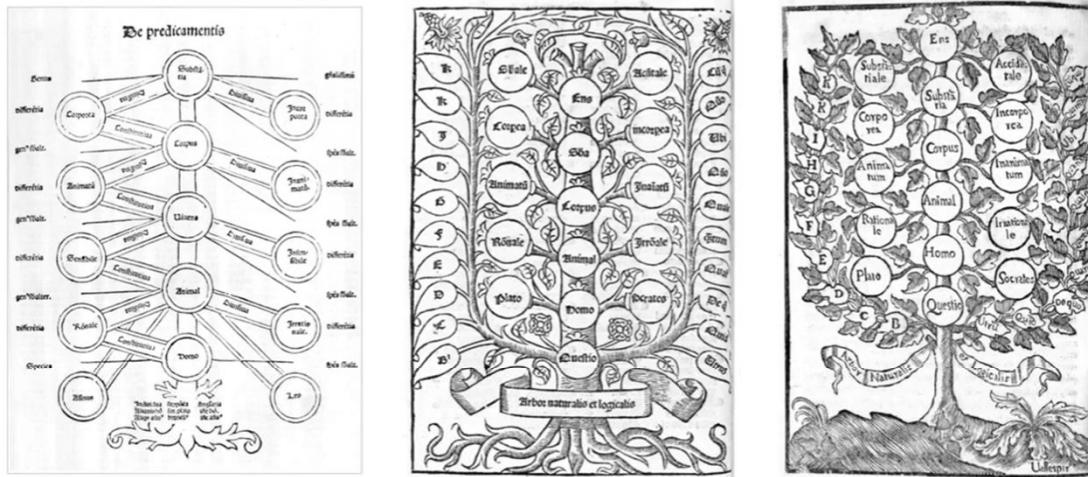
Generative AI

Knowledge Graphs, Graph RAG

Knowledge Graphs, Graph RAG

- Knowledge Graphs
- Graph Database (Neo4j)
- Cypher query language
- Graph RAG / KG RAG

Knowledge Graphs have been a key part of mapping data, throughout human history



Porphyrian Tree (oldest Tree of Knowledge)

A visual history of human knowledge

2,077,878 plays | Manuel Lima | TED2015 • March 2015

<https://www.youtube.com/watch?v=BQZKs75RMqM>

https://www.ted.com/talks/manuel_lima_a_visual_history_of_human_knowledge?language=en

Graph Theory origins

1736 – Leonhard Euler

Which route would allow someone
to cross all 7 bridges



without
crossing

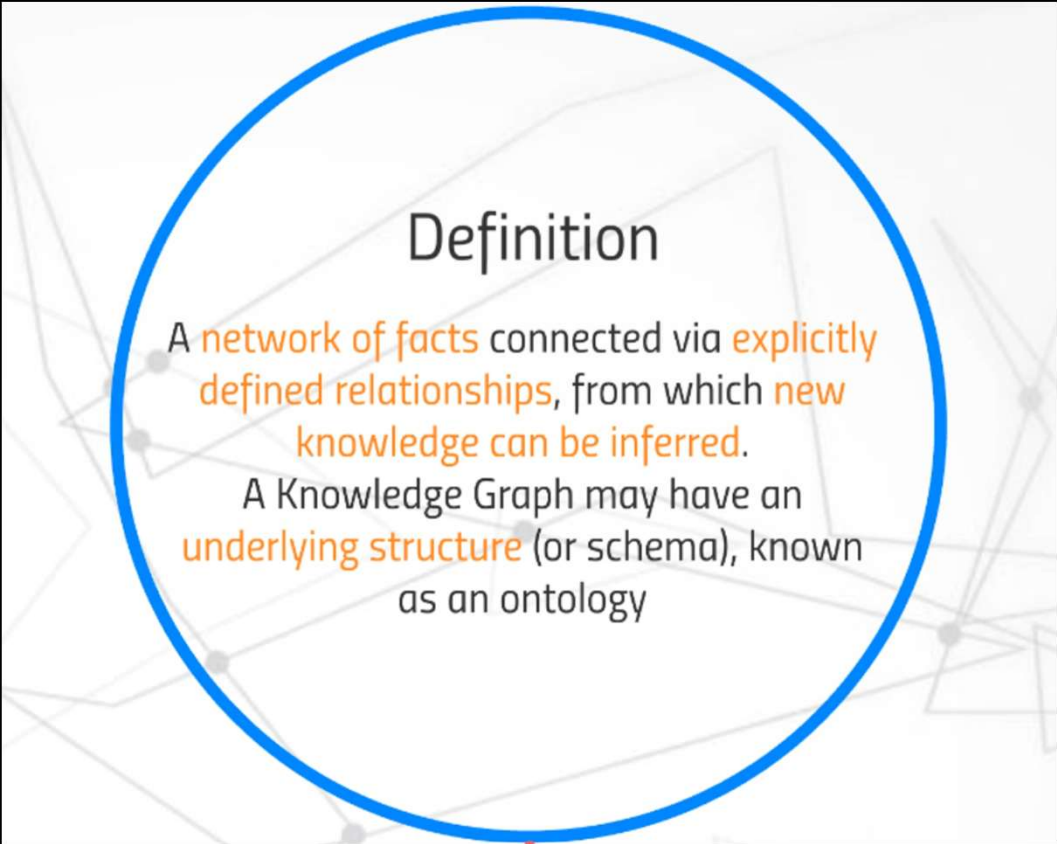
any of them more than once?

[Seven Bridges of Königsberg – Wikipedia](#)

[How the Königsberg bridge problem changed mathematics - Dan Van der Vieren](#)

<https://www.youtube.com/watch?v=ycRuO-u6rt8>

Knowledge Graph - definition

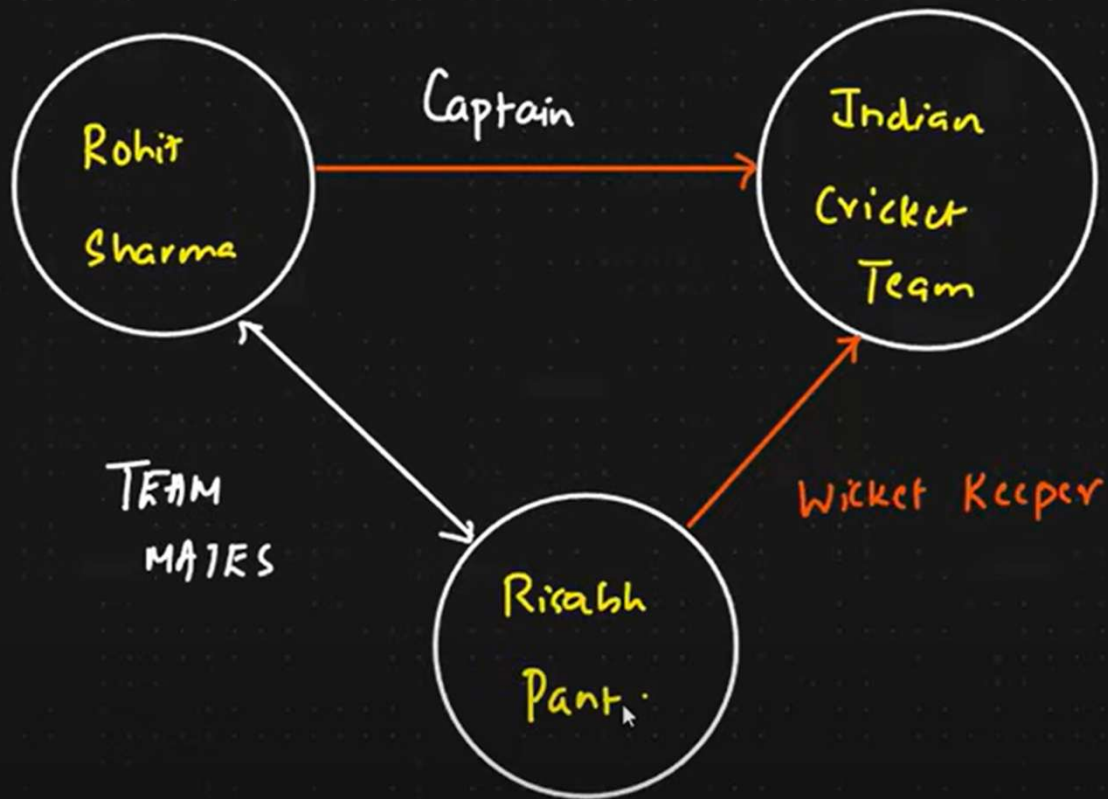


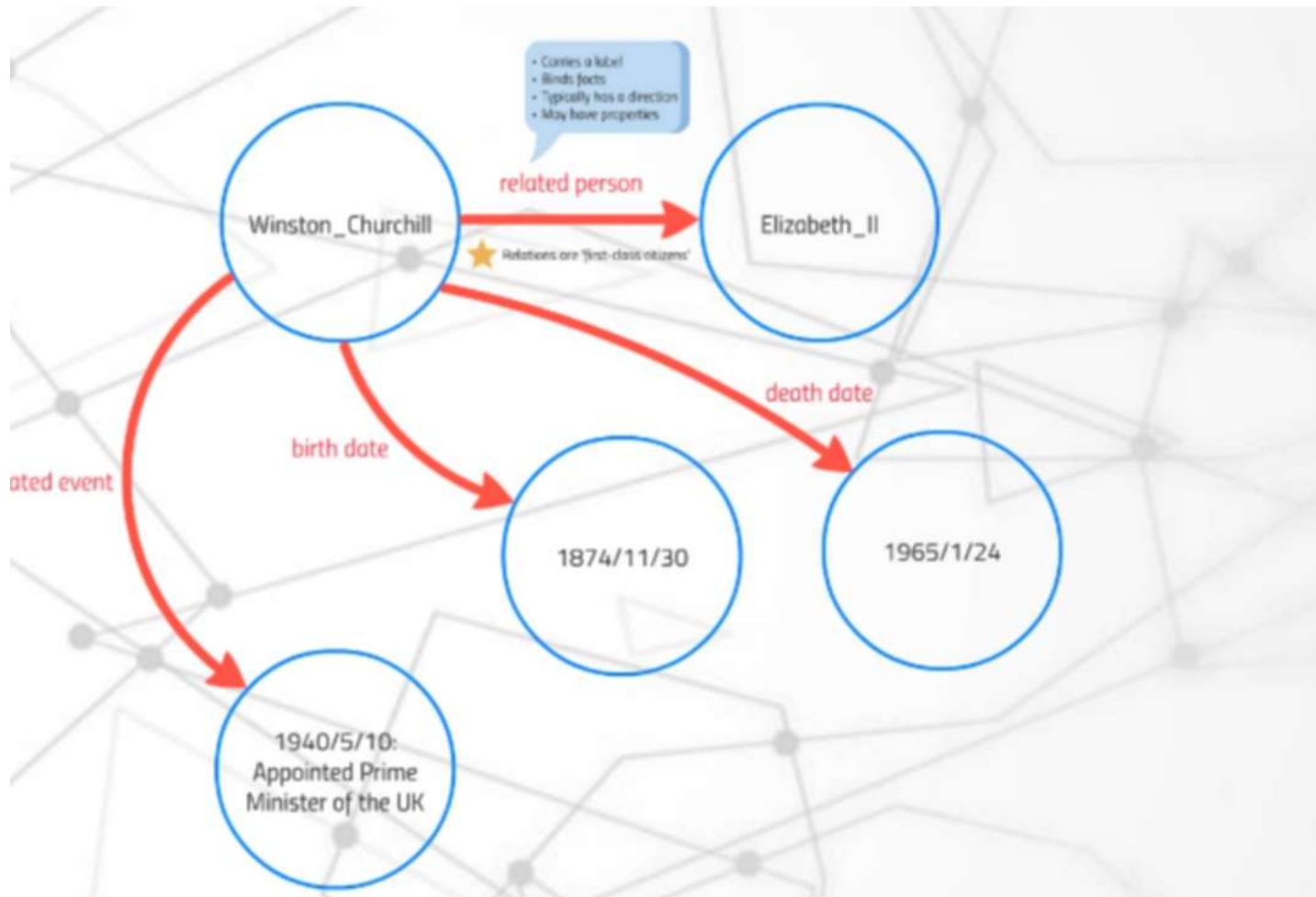
Definition

A **network of facts** connected via **explicitly defined relationships**, from which **new knowledge can be inferred**.

A Knowledge Graph may have an **underlying structure** (or schema), known as an ontology

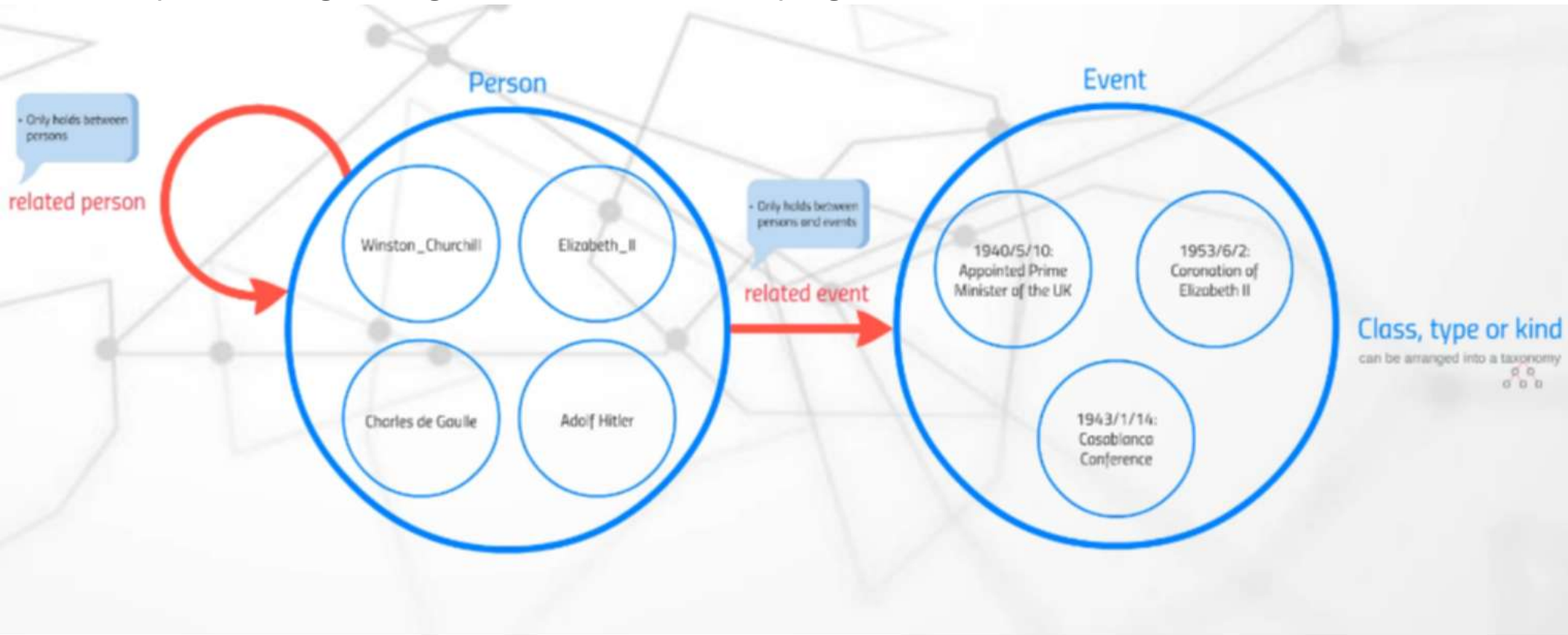
Knowledge Graph – nodes and relations





Ontology

A blueprint for organizing facts, relations. Grouping, relations and rules



Key components of a knowledge graph (a.k.a. KG)

- Nodes
 - e.g., citius:COMPANY (Node label: COMPANY, Node: citius)
- Relationships (Edges)
 - e.g., r1:FRIENDS_WITH (Relationship Type: FRIENDS_WITH, Relation: r1)
- Properties
 - Both nodes and relations can have properties, which are key-value pairs.
For e.g. a Person node p1 has properties name="Vishal", age="30"

Node

A node is an instance that represents a specific entity or concept, and it carries one or more labels that describe its type.

A **node** is an **instance** with:

- one or more **labels** (types / classes)
- properties (attributes)

Example:

```
cypher
```

```
(p1:Patient {patient_id:'P001', name:'Asha Mehta'})
```

Here:

- **Patient** → node label (class)
- **p1** → node variable referring to a specific instance
- this node = an **object**

Nodes (instances)

```
text
```

```
p1:Patient
```

```
p2:Patient
```

```
e1:Encounter
```

```
d1:Diagnosis
```

```
m1:Medication
```

Relationship

A relationship is an instance that connects two nodes and carries a relationship type describing how they are related.

A **relationship** is also an **instance**, with:

- a **relationship type** (label)
- optional properties

Example:

cypher

```
(p1)-[:HAS_DIAGNOSIS]->(d1)
```

Here:

- `HAS_DIAGNOSIS` → relationship type (verb)
- this edge = an instance of that relationship

Relationships (instances)

text

```
(p1)-[r1:HAD_ENCOUNTER]->(e1)
```

```
(e1)-[r2:HAS_DIAGNOSIS]->(d1)
```

```
(e1)-[r3:PRESCRIBED]->(m1)
```

Property

Attribute of a node or relationship

A **property** is a **key–value attribute** stored on a node or relationship.

- Think: **fields**
- Node property examples:
 - `Patient.name` , `Encounter.date` , `Diagnosis.icd10` , `Medication.rxnorm` , `ClinicalNote.text`
- Relationship property examples (when needed):
 - `MENTIONS.score` , `PRESCRIBED.dose`

Aligns with OOP

OOP

java

```
patient1.hasDiagnosis(diagnosis1);
```

Graph

cypher

```
(patient1)-[:HAS_DIAGNOSIS]->(diagnosis1)
```

Label shortcuts are sometimes used directly, specially for relations, as seen above

Working with graph databases – which language?

- Cypher
 - Relationships (Edges)
 - Directional, e.g., FRIENDS_WITH
- Gremlin
 - Used to group nodes of a certain type, for e.g., all person nodes can have label Person
- Properties
 - Both nodes and relations can have properties, which are key-value pairs. For e.g. a Person node may have name="Vishal", age="30"
- Relationship Type
 - Type of relationship, for e.g. WORKS_FOR

Knowledge Graph example use-case 1

The "Hidden Champion" Discovery (Sales & Marketing)

Sales Rep asks, "Why should I call this lead?"

Knowledge Graph assisted AI explains the deep relationship path.

Instead of just looking at the industry or size of these customers, the KG analyzes deep relations:

- Common Board Members: Do these companies share a specific board member?
- Technology Stack Synergies: Do they all use a specific third-party ERP module that integrates perfectly with your suite?
- Executive Peer Groups: Are the decision-makers at these companies part of the same professional networking circle?

The "Aha!" Moment: The KG discovers a "Hidden Champion" prospect that matches all these non-obvious relationship patterns, even though they aren't in the same industry as your other customers.

Knowledge Graph example use-case 2

The "Supply Chain Fragility" Audit (Procurement)

Scenario 1: The "Shared Tier-3" Bottleneck (Semiconductors)

You source electronic control units (ECUs) from two different "Tier-1" suppliers: Bosch and Continental. On paper, your risk is diversified because you have two independent vendors in different locations.

KG shows that both Bosch and Continental source their microchips from a "Tier-2" provider, TSMC. So far so good. Now KG reveals a deep relation: TSMC relies on a single "Tier-3" refinery, Neon-Tech Corp, for high-purity neon gas used in lasers.

If Neon-Tech Corp shuts down due to a regional crisis, both of your "independent" Tier-1 suppliers fail simultaneously.

A similar scenario: Any geo-related issue such as a block in Suez Canal can be analyzed for impact

Knowledge Graph example use-case 3

The "Conflict of Interest" Red-Flagger (Compliance)

The Scenario: You are about to award a \$10M contract to "Alpha Consulting."

KG scans the relationship web:

Conflict 1:

Entity Linkage: Alpha Consulting is a subsidiary of Beta Holdings.

The Conflict: Beta Holdings recently hired the spouse of your VP of Procurement.

Conflict 2:

The Discovery: The KG finds a "circular relationship" where Alpha Consulting's lead negotiator was previously the auditor for your firm's internal finance team.

The "Aha!" Moment:

The graph highlights a high-risk conflict of interest that spans across different legal entities, departments, and historical employment records—connections that a table-based search would completely miss.

Cypher – pattern matching language from Neo4j

Milestone	What happened	Why it matters
2012 Neo4j 2.0	Cypher introduced as a proprietary, SQL-inspired pattern-matching language.	Quickly became the best-known “property-graph” query language.
2015 openCypher	Neo4j opened the language spec and grammar under the LDBC consortium.	Vendors such as RedisGraph, Memgraph, SAP HANA Graph and Apache AGE adopted it. AWS Documentation
2024 ISO/IEC 39075 GQL	The first <i>formal</i> international Graph Query Language standard was published. It borrows syntax and semantics from openCypher, PGQL and G-CORE but is not identical to any of them. ISO Graph Database & Analytics	

Adopted by Amazon Neptune (openCypher), TigerGraph, besides of course, neo4j

Gremlin – pattern matching language from Neo4j

- Conceived in 2009 by **Marko A. Rodriguez**
- Released through the open-source **Apache TinkerPop** project, which he co-founded.
- TinkerPop positions Gremlin as a “**graph traversal machine and language**,” analogous to how the JVM hosts Java programs.

Adopted by Amazon Neptune, JanusGraph, Azure Cosmos DB

Popular graph databases

Name	Strengths	Weaknesses	Licence / Deployment model	Query-language support
Neo4j (Neo4j Inc)	• Mature ecosystem, rich tooling (APOC, Bloom, drivers) • Powerful graph algorithms & GDS library • Strong community & training resources	• Some features in enterprise-only edition (clustering, RBAC) • High memory footprint for very large graphs • GPL v3 for Community can deter closed-source redistribution	Community Edition — GPL v3; Enterprise — commercial subscription (Graph Database & Analytics)	Native Cypher (plus preview ISO GQL in v5.x)
Amazon Neptune	• Fully-managed, auto-patched & backed-up • Choice of property-graph and RDF models • Deep integration with the AWS stack	• Vendor lock-in to AWS • Separate endpoints for PG/RDF • Pricing driven by instance size + I/O	Closed-source, AWS-managed PaaS (AWS Documentation)	openCypher, Gremlin, SPARQL (AWS Documentation)
ArangoDB (ArangoDB Inc)	• Multi-model (documents + KV + graphs) in one engine • Flexible JOIN-like traversals via AQL • Good built-in clustering & sharding	• Smaller graph-specific ecosystem • Licence changed to BSL 1.1 for future versions (cloud-hosting restriction)	Apache 2.0 ≤ v3.11; BSL 1.1 (falls back to Apache 2.0 after 4 yrs) from v3.12+ (ArangoDB, ArangoDB)	Native AQL

Popular graph databases (contd.)

Name	Strengths	Weaknesses	Licence / Deployment model	Query-language support
TigerGraph (TigerGraph Inc)	• High-performance distributed engine tuned for real-time analytics • Parallel, Turing-complete GSQL language • Built-in graph data-science and vector search	• Proprietary closed source (free dev edition only) • Learning curve vs SQL/Cypher • Smaller community than Neo4j	Proprietary, node-locked licence; free single-node Dev edition (TigerGraph Documentation)	GSQL (plus optional openCypher subset) (TigerGraph Documentation)
JanusGraph (Linux Foundation)	• 100 % open-source, vendor-neutral • Back-end pluggable (Cassandra, HBase, Bigtable, etc.) • Massively scalable with external storage & index layers	• Requires external DB + search engine (ES/Solr/Lucene) • Operational complexity vs all-in-one stores • No native query DSL beyond Gremlin	Apache 2.0 (JanusGraph)	Gremlin (TinkerPop); SPARQL via plugin (JanusGraph)
Azure Cosmos DB (Gremlin API)	• Globally distributed with multi-region writes • Millisecond P99 latency SLA • Multi-model (graph, docs, KV, column) under one service	• RU/s cost model can be hard to predict • Gremlin feature set lags behind on-prem engines • Closed-source cloud service	Proprietary, fully-managed PaaS by Microsoft Azure (Microsoft Azure)	Gremlin API (plus SQL, Mongo, Table, Cassandra APIs) (Microsoft Azure)

Example – nodes, relations, properties creation using Cypher

```
CREATE (p:Person {name: "Alice", age: 30, city: "New York", email: "alice@example.com"})
```

```
CREATE (c:Company {name: "Acme Corp", location: "San Francisco", industry: "Technology"})
```

```
CREATE (p)-[r:WORKS_FOR {since: 2015, role: "Software Engineer", department: "R&D"}]->(c)
```

```
CREATE (p)-[r:FAN_OF]->(c)
```

```
CREATE (p:Person {name: "Bob", age: 30, city: "New York", email: "alice@example.com"})
```

```
CREATE (p)-[r:WORKS_FOR {since: 2015, role: "Software Engineer", department: "R&D"}]->(c)
```

- How many nodes?
- How many Node Labels?
- How many relationship types?
- How many relations?
- Nodes: 3, Node labels: 2 (Person, Company)

Cypher Queries

// Show all nodes and relationships

```
MATCH (n)-[r]->(m)
```

```
RETURN n, r, m
```

// Clear graph

```
MATCH (n) DETACH DELETE n
```

Language families used by popular graph databases

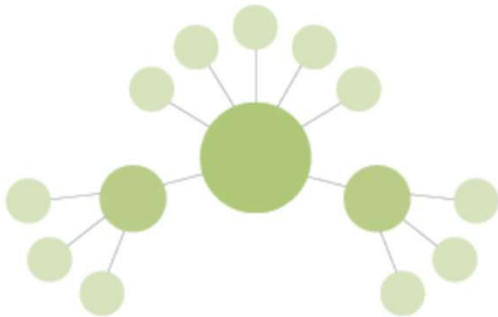
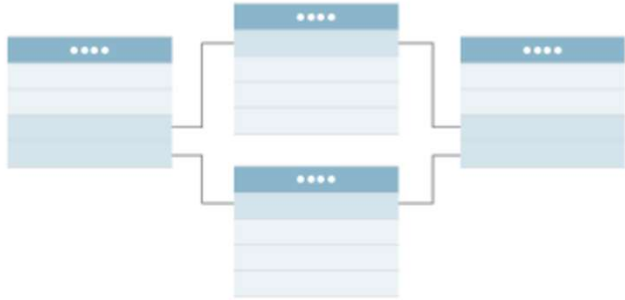
Language family	Typical database engines ¹	Notes
openCypher / Cypher	Neo4j, RedisGraph, Memgraph, SAP HANA Graph, Amazon Neptune (preview), NebulaGraph, Apache AGE	De-facto “property-graph” lingua franca, but implementation details diverge.
Gremlin (Apache TinkerPop)	JanusGraph, Azure Cosmos DB (Gremlin API), Amazon Neptune, OrientDB, DataStax Enterprise Graph	Imperative traversal style; vendor-neutral framework.
SPARQL (W3C standard)	GraphDB, Blazegraph, Virtuoso, Stardog, Neptune (RDF mode)	Targets RDF triple stores rather than property graphs.
GSQL	TigerGraph	Declarative, graph-native analytics extensions.
AQL	ArangoDB	Multi-model (documents + graphs) query language.
PGQL	Oracle PGX, Oracle Database 23c	SQL-like, one of the inputs to ISO GQL.
SQL/PGQ (part 16 of SQL:2023)	Rolling into mainstream relational DBs (e.g., PostgreSQL prototypes, Oracle 23c)	Lets you run property-graph pattern matching <i>inside</i> SQL. Peter Eisentraut ISO

Demo with neo4j

If firewall issues, use ssc to bypass certificate check

NEO4J_URI=neo4j+ssc://5df217b6.databases.neo4j.io

Graph Databases Vs RDBMS

	Graph database	Relational database
FORMAT	Nodes and edges	Tables with rows and columns
RELATIONSHIPS	Considered data, represented by edges between nodes	Related across tables, established using foreign keys between tables
COMPLEX QUERIES	Run quickly and do not require joins	Require complex joins between tables
TOP USE CASES	Relationship-heavy use cases, including fraud detection and recommendation engines	Transaction-focused use cases, including online transactions and accounting
EXAMPLE		

Neo4j Property Graph Model

Database information

Nodes (2)

 **Company**  **Person**

Relationships (1)

 **WORKS_FOR**

Property keys

age **city** **data** **department** **email** **id**

industry **location** **name** **nodes**

relationships **role** **since** **style**

visualisation

neo4j\$

```
neo4j$ MATCH (n) WHERE n.city IS NOT NULL RETI
```

Table RAW

	entity	city
1	"node"	"New York"

Started streaming 1 record after 48 ms and comp

```
1 MATCH (n)
2 WHERE n.city IS NOT NULL
3 RETURN DISTINCT "node" as entity, n.city AS city
4 LIMIT 25
5 UNION ALL
6 MATCH ()-[r]-()
```

Nested Queries

Scenario

After John Doe's X-Ray procedure in a specific visit, show me any diagnoses recorded within 30 days of that procedure.

```
// 1) Find the initial X-Ray visit for John Doe.
MATCH (p:Patient { name: "John Doe" })-[:UNDERWENT_VISIT]->(initialVisit:Visit)
      -[:HAS_PROCEDURE]->(xray:Procedure { type: "X-Ray" })
WHERE initialVisit.dateOfVisit = date("2025-01-15") // or any relevant date filter

// 2) Use WITH to pass forward the patient and the initial visit.
WITH p, initialVisit

// 3) From the same patient, find subsequent visits that have any diagnosis.
//     Only keep those visits that occurred within 30 days after the X-Ray visit.
MATCH (p)-[:UNDERWENT_VISIT]->(followupVisit:Visit)-[:HAS_DIAGNOSIS]->(diag:Diagnosis)
WHERE followupVisit.dateOfVisit > initialVisit.dateOfVisit
      AND duration.between(date(initialVisit.dateOfVisit), date(followupVisit.dateOfVisit)).days < 30

// 4) Return the details of those follow-up diagnoses
RETURN diag.name AS diagnosisName, followupVisit.dateOfVisit AS diagnosisDate
ORDER BY followupVisit.dateOfVisit
```

Nested Queries

Scenarios

After John Doe's X-Ray procedure in a specific visit, show me any diagnoses recorded within 30 days of that procedure.

Find any additional visits for the same patient after a medication was prescribed.

Show me all diagnoses that eventually lead to a referral to a specific doctor.

Within a certain time window, find subsequent procedures or medications.

Shortest Path Queries

Scenarios

Show me the shortest path between Dr. Alice (a Doctor) and Nurse Bob (a Nurse), and how many edges (hops) are between them?

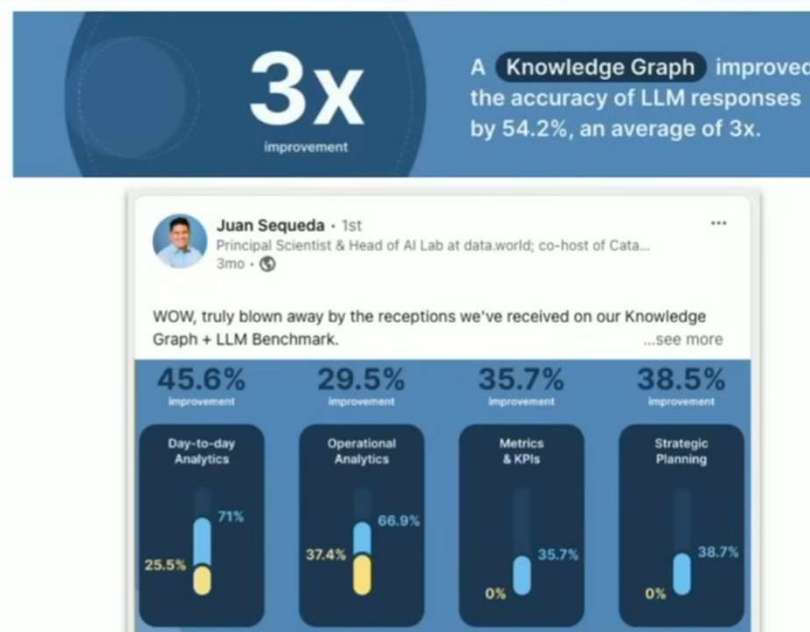
How are these two doctors connected via their referrals, shared departments, or patient handoffs?

Which chain of people (nurses/doctors) did a patient pass through before reaching the specialist?

Knowledge graph - benefits

① Higher Accuracy

The clearest driver of GraphRAG adoption amongst users is *higher accuracy*.



<https://www.youtube.com/watch?v=knDDGYHnnSI>

Knowledge graph - benefits

② Easier Development*

The second reason we hear people choose GraphRAG over vector-only RAG is *easier development*... once they've pushed through the initial learning curve.

* Weirdly, ease of development [or lack thereof] is also one of the stumbling blocks!

Feedback from an AI Engineer

I kinda replicated the same action-based cache already in place for Pinecone but thanks to the graph nature of Neo4J most of the operations better results:

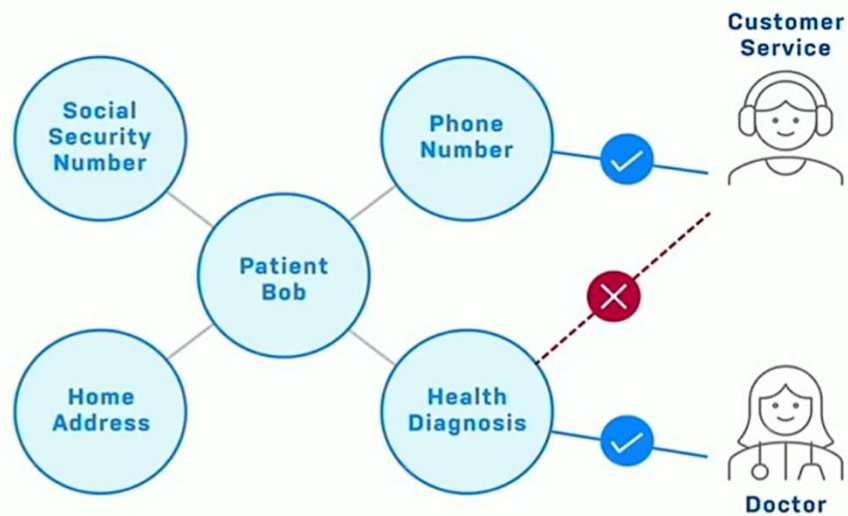
- thanks to the `Neo4j graph data science` plugin we can store embeddings and calculate cosine similarity at the database level
- getting related actions is as simple as following the relationships between nodes

Caveat: creating KG takes work, but once created ease of development is a big plus

Knowledge graph - benefits

③ Explainability & Governance

Finally, in particular central IT & business decision makers cite ease of explainability and governance.



Leveraging knowledge graph for chronological data

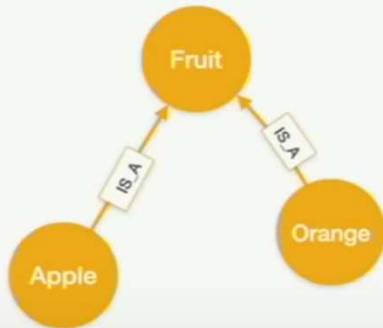
Usecase: medical records summarization in a chronological order

Knowledge graphs can be very useful:

- Date or timestamp can be a **property** on the MEDICAL_RECORD node
You can then use the ORDER_BY in cypher query to retrieve the records chronologically
- Temporal relationships
You can model a :NEXT_EVENT relation time to stitch the events to create a chain of events
- Temporal Queries will be fast
MATCH (r:MedicalRecord)
RETURN r
ORDER BY r.date ASC

KG Vs regular RAG

Representation



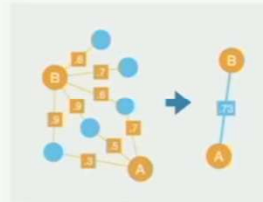
Representation

Apple	[0.2435, 3.7652, 0.00234, 456.66, ...]
Orange	[115.124, 29.7652, 4.2131, 2.431, ...]
Fruit	[0.0035, 17.661, 0.0113, 11.4566, ...]

Similarity calculation

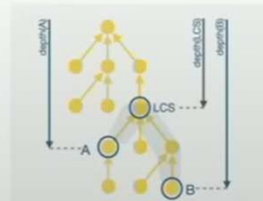
Structural

- ▶ Node similarity
- ▶ Overlap
- ▶ Jaccard



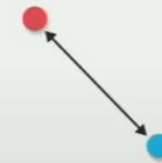
Taxonomy based

- ▶ Path
- ▶ Leacock-Chodorow
- ▶ Wu-Palmer

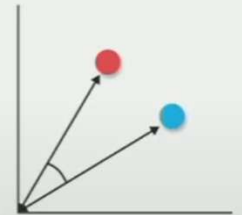


Similarity calculation

Euclidean

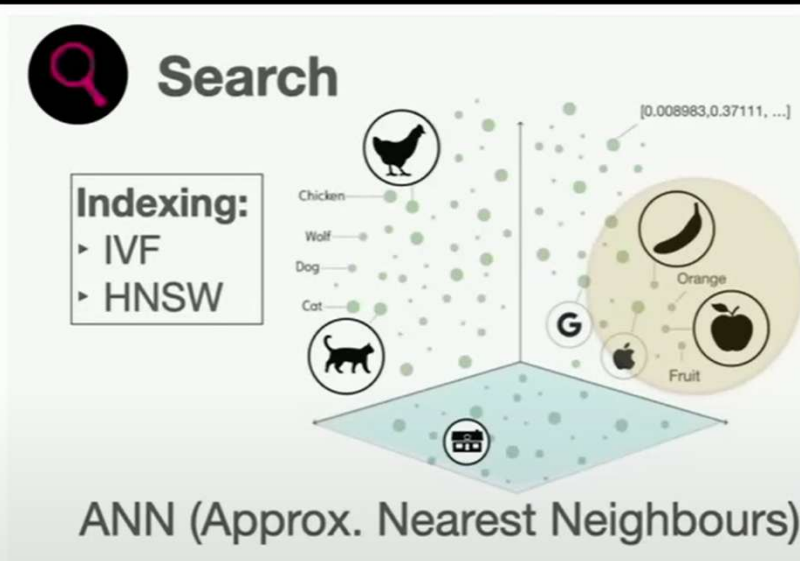
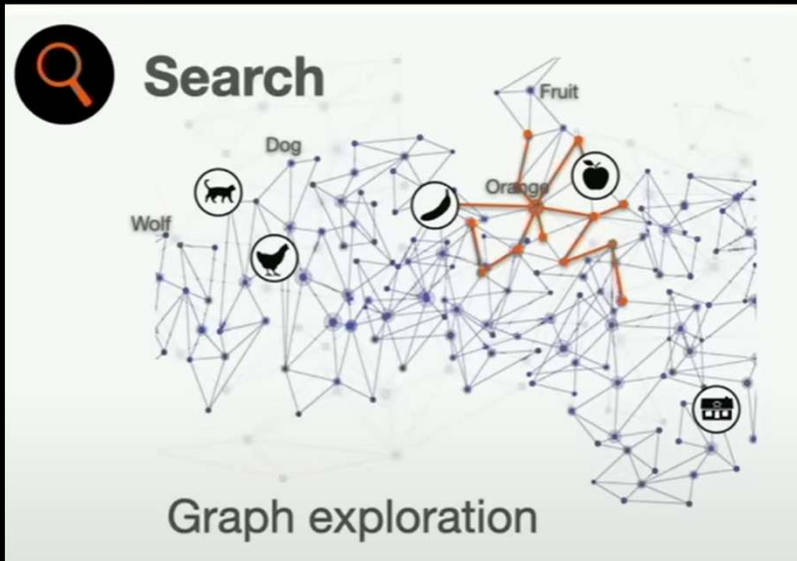


Cosine



Vector distance based

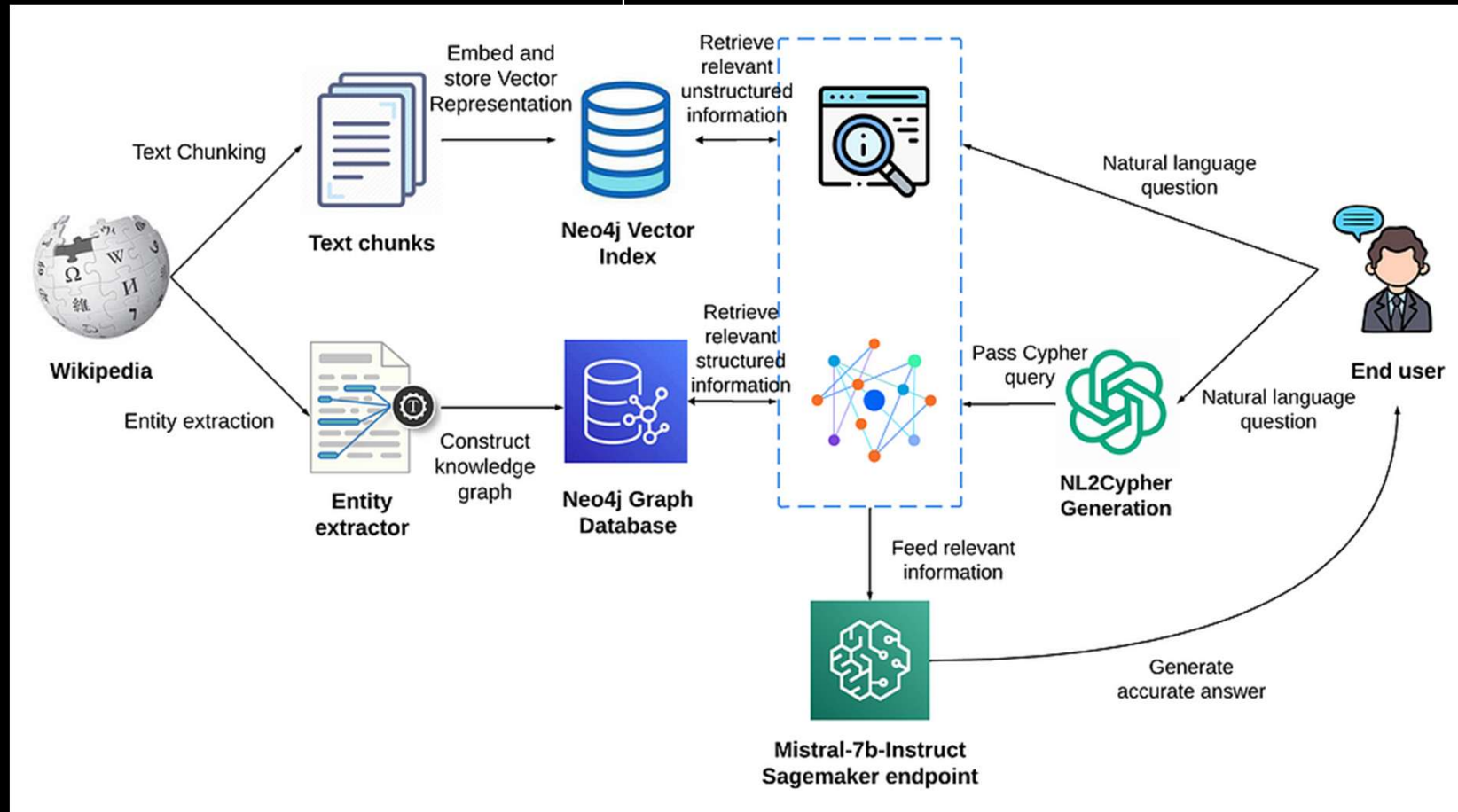
KG Vs regular RAG



Managing unstructured text and KG in neo4j

Indexing

Retrieval



Storage Architecture

Document Store (optional third component):

- Store original documents and metadata
- Maintain source references and provenance
- Examples: MongoDB, Elasticsearch, PostgreSQL, S3

Vector Database (for descriptive text):

- Store text chunks, paragraphs, sections
- Capture semantic meaning through embeddings

Graph Database (for structured knowledge):

- Store entities, relationships, and their metadata
- Maintain explicit connections between concepts
- Enable traversal and reasoning

Indexing steps with Graph Database

- Text extraction from documents (PDF for example)
- Entities extraction
- Relationships extraction
- Graph construction and storage

Graph creation tools and libraries

- Neo4j Graph Builder
- LLM
- Python libraries

Schema enforcement in neo4j

Neo4j out of box does not enforce any fixed schema unlike SQL/RDBMS.

So without guardrails, this can lead to “property soup” (patientId vs patient_id), silent typos, inconsistent labels/relationship types, and messy queries.

Schema enforcement in neo4j – best practices

1) Enforce uniqueness + required identifiers with constraints

This prevents duplicate nodes and forces a consistent key to exist (at least for the identity you care about)

- **Uniqueness constraint** (best practice for every main entity):

cypher

```
CREATE CONSTRAINT patient_id_unique IF NOT EXISTS  
FOR (p:Patient) REQUIRE p.patient_id IS UNIQUE;
```

- **Existence (required property) constraints** (Neo4j supports property existence constraints for nodes/relationships in modern versions):

cypher

```
CREATE CONSTRAINT patient_id_exists IF NOT EXISTS  
FOR (p:Patient) REQUIRE p.patient_id IS NOT NULL;
```

Validate payloads before writing (app layer / ETL guardrails)

- Define a **canonical schema** in your ingestion layer (Pydantic / JSON Schema / Great Expectations-like checks).
- Reject or quarantine records with:
 - unknown fields
 - wrong types
 - missing required fields
 - enum violations (e.g., gender not in {"M","F","O","U"})

Example approach (conceptual):

- Pydantic models per node type (PatientCreate, EncounterCreate)
- `extra="forbid"` so unknown keys fail fast
- normalize names (snake_case), trim whitespace, standardize codes, etc.

This is the single most effective way to prevent spelling mistakes.

Add schema monitoring queries

- Find unexpected properties for a label

cypher

```
MATCH (n:Patient)
UNWIND keys(n) AS k
RETURN k, count(*) AS cnt
ORDER BY cnt DESC;
```

Compare returned keys against your allowed list; alert on unknown ones.

- Find near-duplicate keys / typos (simple heuristic)

cypher

```
MATCH (n:Patient)
UNWIND keys(n) AS k
WITH collect(DISTINCT k) AS ks
RETURN ks;
```

Then in your app/script, compute similarity (`patient_id` vs `patinet_id`) and flag.

Schema design – nodes, relationships

```
(:Patient)-[:HAS_DIAGNOSIS]->(:Disease)  
(:Patient)-[:VISITED]->(:Hospital)
```

```
(:Doctor)-[:TREATED {since:2023, outcome:"Recovered"}]->(:Patient)
```

```
(:Disease)-[:IS_A]->(:ChronicDisease)  
(:Diabetes)-[:IS_A]->(:MetabolicDisorder)
```

```
(:Patient)-[:HAS_EVENT]->(:Admission)  
(:Admission)-[:AT_TIME]->(:Date)
```

admission_date should be a node or property?

admission_date: node or property?

Requirement	Property	Date Node
Filter by date	✓	✗
Sort by time	✓	✗
Patient journey timeline	✓	✗
Time hierarchy (day → month → year)	✗	✓
Cross-event aggregation	✗	✓
Seasonal / trend analysis	✗	✓
Graph algorithms over time	✗	✓
Graph + Star hybrid parity	✗	✓

Which one is better?

```
(:Patient)-[:HAS_AGE]->(:Age {value: 45})  
(:Patient)-[:HAS_GENDER]->(:Gender {value:"M"})  
(:Patient)-[:HAS_CITY]->(:City {value:"Pune"})
```

```
(:Patient {patient_id:"P123", age:45, gender:"M", city:"Pune"})
```

Which one is better?

```
(:Patient {diagnoses:["E11","I10"], doctors:["D1","D8"], hospitals:["H2"]})
```

```
(:Patient)-[:HAS_DIAGNOSIS {first_seen:"2025-09-11", severity:"moderate"}]->(:Disease {code:"E11"})  
(:Doctor)-[:TREATED {date:"2025-09-11"}]->(:Patient)
```

Show me all patients diagnosed with E11 disease

P1: [e11, i10], p2[e3, e4].....

Match :p where disease.code="E11"

What's wrong with this?

```
(:Patient)-[:VISITED]->(:Hospital)  
(:Patient)-[:HAS_DIAGNOSIS]->(:Disease)
```

Direct edges that hide time/episodes

No timeline, No per-visit context, Can't answer: "what diagnosis at which visit?"

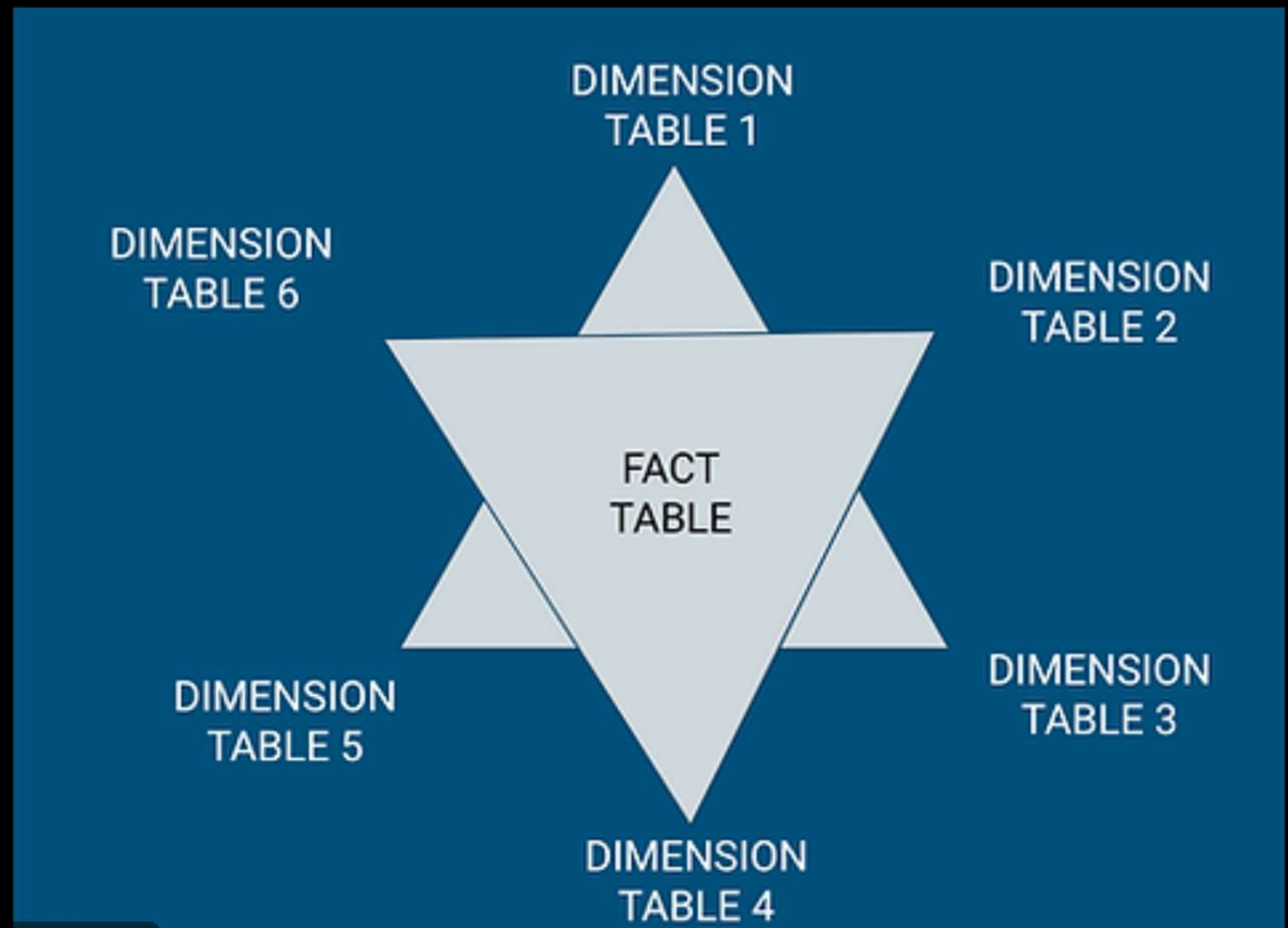
```
(:Patient)-[:HAD_ENCOUNTER]->(:Encounter {encounter_id:"E9001", start_ts:...})  
(:Encounter)-[:AT_FACILITY]->(:Facility)  
(:Encounter)-[:HAS_DIAGNOSIS]->(:Disease)  
(:Encounter)-[:PRESCRIBED]->(:MedicationOrder {dose:"500mg", freq:"BID"})
```

Event centric, fact as node pattern

STAR Schema

Structure

- One central **Fact Table**
- Surrounded by **Dimension Tables**
- Looks like a ★ (hence the name)



STAR Schema example

Fact Table

diff

FACT_ADMISSIONS

- admission_id
- patient_id
- hospital_id
- doctor_id
- date_id
- cost
- length_of_stay

Dimension Tables

SCSS

DIM_PATIENT(patient_id, age, gender)
DIM_DOCTOR(doctor_id, specialty)
DIM_HOSPITAL(hospital_id, city)
DIM_DATE(date_id, year, month)

- ✓ Extremely fast aggregations
- ✓ BI-tool friendly
- ✗ Poor for relationship reasoning

In graphs:

- Fact table → Event node
- Dimensions → Entity nodes
- Foreign keys → Relationships

STAR Schema vs Graph native design

Aspect	Star Schema (RDBMS)	Graph Schema
Primary goal	Aggregation	Relationship reasoning
Structure	Tables + joins	Nodes + relationships
Query style	GROUP BY	Traversals
Time-based analysis	Excellent	Good (with event nodes)
Multi-hop questions	Poor	Excellent
Schema rigidity	High	Flexible

STAR Schema vs Graph native design

Use Star Schema When

- BI dashboards
- KPI reporting
- Monthly / quarterly metrics
- Power BI / Tableau / SQL workloads

Operational Data → Graph **DB** (reasoning)
Analytical Data → Star **Schema** (aggregation)

Use Graph Schema When

- Patient journeys
- Doctor–disease–treatment reasoning
- Root cause analysis
- Fraud, risk, or clinical pathways
- Graph RAG + LLM reasoning

Star schema asks: “How much / how many?”
Graph schema asks: “How is X connected to Y—and why?”

Other schema patterns

1) Snowflake schema (dimensional)

- Like star, but **dimensions are normalized** into sub-dimensions.
- Example: `dim_facility` → `dim_city` → `dim_state`.
- Pros: less redundancy, better governance. Cons: more joins, slower BI if overdone.

2) Galaxy / Fact constellation

- **Multiple fact tables share dimension tables.**
- Example: `fact_encounter`, `fact_lab_result`, `fact_med_admin` all share `dim_patient`, `dim_date`, `dim_facility`.
- Pros: models multiple business processes cleanly; common in enterprise warehouses.

3) 3NF / OLTP normalized model

- Classic transactional design: minimize redundancy using **3rd Normal Form**.
- Example: `Patient`, `Doctor`, `Encounter`, `Diagnosis`, `LabResult` with strict foreign keys.
- Pros: integrity + update correctness. Cons: heavy joins for analytics.

Other schema patterns (contd.)

4) Data Vault (Hub–Link–Satellite)

- Designed for enterprise integration & auditability:
 - **Hubs** = business keys (PatientID)
 - **Links** = relationships (Patient–Encounter)
 - **Satellites** = attributes + history (patient address changes over time)
- Pros: great for lineage/history; Cons: complex for beginners, BI layer needed.

5) Wide table / denormalized “One Big Table”

- Flattened table for fast reads in some analytics cases.
- Pros: simple queries. Cons: duplication, harder updates, data quality risk.

6) Event sourcing / append-only ledger tables

- Store changes as events instead of updating rows.
- Pros: perfect history & replay. Cons: queries require reconstruction/materialized views.

Graph embeddings and semantic relationships (neo4j GDS)

Why graph embeddings?

Semantic similarity from structure + context

Graph Data Science concepts:

Projection, in-memory graph, write-back properties

Algorithms:

FastRP (fast, reliable for training)

Node2Vec (optional, random-walk intuition)

Use-cases:

Similar diagnoses / medications (cosine similarity)

Similar encounters & recommendation

A graph embedding is learned from:

- Who is connected to whom
- Paths, hops, neighborhoods
- Shared neighbors
- Graph roles (hub, bridge, leaf)

It does NOT need text.

- Patient A and Patient B
- have different notes and no shared words
- but same diagnoses + meds via similar encounter paths

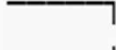
➡ Their **graph embeddings** will be similar

Even if text embeddings say they're unrelated.

Graph embeddings vs Text Embeddings

Graph Embeddings

Learns a vector for each node so that:
Nodes with connections to similar nodes
get similar vectors

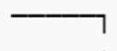
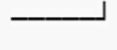
```
Patient_A — Encounter — Hypertension —   
Patient_B — Encounter — Hypertension —  (close)
```

```
Patient_C — Encounter — Diabetes — Metformin
```

Text Embeddings

A vector DB embedding represents the semantic
meaning of content

Created by models like:
Sentence-BERT
OpenAI embeddings
MiniLM, etc.

```
"high blood pressure" —   
"hypertension" —  (close)
```

Graph embeddings vs Text Embeddings

Question

"Show patients similar to P001 with hypertension"

Graph embedding does:

- Finds patients with:
 - same diagnosis nodes
 - same medication nodes
 - similar encounter patterns

Text embedding does:

- Ranks notes mentioning:
 - "BP high"
 - "hypertension"
 - "uncontrolled BP"

Extracting entities and relationships from text

- Baseline NER (spaCy) + why domain models matter (healthcare)
- Dictionary / terminology-driven NER from KG (high precision)
- Entity Linking strategies:
 - exact match (codes, canonical names)
 - **fuzzy match** (RapidFuzz)
 - **semantic match** (Sentence-Transformers) for longer phrases
- Writing back to Neo4j:
 - ClinicalNote nodes, MENTIONS edges with method, score, mention
- Real-world complications:
 - negation (“no diabetes”), abbreviations, disambiguation, confidence, HITL

Outcome: extract entities from text and link them to canonical graph nodes with traceable confidence.

Coordination strategies – Indexing

Linked storage pattern

```
# Vector chunk with graph references
vector_chunk = {
    "id": "chunk_123",
    "text": "The research team led by Dr. Smith discovered...",
    "embedding": [0.1, 0.2, ...],
    "entities": ["Dr. Smith", "research_project_X"], # Links to graph
    "chunk_position": {"page": 5, "paragraph": 2}
}

# Graph entity with vector references
graph_entity = {
    "id": "Dr_Smith",
    "type": "Person",
    "properties": {...},
    "text_chunks": ["chunk_123", "chunk_456"] # Links to vector DB
}
```

Coordination strategies – Retrieval

Phase 1: Multi modal retrieval

```
def hybrid_retrieve(query):  
    # 1. Vector similarity search  
    similar_chunks = vector_db.similarity_search(query, k=10)  
  
    # 2. Graph traversal from query entities  
    query_entities = extract_entities(query)  
    related_subgraph = graph_db.traverse_from_entities(query_entities)  
  
    # 3. Cross-reference and rank  
    combined_results = merge_and_rank(similar_chunks, related_subgraph)  
    return combined_results
```


Coordination strategies – Retrieval

Phase 2: Context assembly

```
def hybrid_retrieve(query):  
    # 1. Vector similarity search  
    similar_chunks = vector_db.similarity_search(query, k=10)  
  
    # 2. Graph traversal from query entities  
    query_entities = extract_entities(query)  
    related_subgraph = graph_db.traverse_from_entities(query_entities)  
  
    # 3. Cross-reference and rank  
    combined_results = merge_and_rank(similar_chunks, related_subgraph)  
    return combined_results
```

Integrated graph retrieval system

Initial Filtering (Keyword search)

A quick keyword search to narrow down a large corpus to a manageable set of candidates.
Such as metadata filtering

Semantic search and ranking

Apply embeddings and cosine similarity to rank the candidates based on their semantic relevance.

Relationship Analysis

Finally, incorporating graph relationships to refine or contextualize the search results, such as ranking based on node connectivity or importance.

Retrieval order and implementation options

Pattern 1: Graph-Anchored Chunks

- Store text chunks that are "anchored" to graph entities:
- Each chunk tagged with entities it discusses
- Graph entities link back to relevant chunks
- Retrieval starts with entities, expands to descriptive content

Retrieval order and implementation options

Query routing

```
def route_query(query):  
    query_type = classify_query(query)  
  
    if query_type == "factual":  
        # Start with graph, expand to text  
        return graph_first_retrieval(query)  
    elif query_type == "descriptive":  
        # Start with vector, add graph context  
        return vector_first_retrieval(query)  
    else:  
        # Balanced hybrid approach  
        return parallel_retrieval(query)
```

Retrieval order and implementation options

Fuse results

```
def fuse_results(vector_results, graph_results):  
    # Score and rank results from both sources  
    # Remove duplicates and contradictions  
    # Organize by relevance and source credibility  
    return unified_context
```